👩‍🏫 Staff Guide · Small Engines Overhaul · Unit 8 Engine Teardown & Rebuild (Level 2)

Level 2 = about 3rd-grade reading. SLE students stay in the class teacher's lesson and use these pieces next to it: same topics, same order, same points.

1. How the pieces line up

Gen ed piece (Canvas: "Weeks 6-9+: engine project (teardown and rebuild), Ring End Gap/Land Clearance Handout, How to grind valves video, Engine Assembly 1, iCADEMY tech bits (engine quick check, rewind starter repair, valve clearance, throttle cable, crankcase sealing, chain, trimmer, blower and cut-off saw service, chaps)")	**Level 2 piece**
Engine project (teardown/rebuild, not written in Canvas)	Teardown & Rebuild lab packet: step-by-step picture checklist
Ring End Gap / Land Clearance Handout	Notes B + Ring Gap steps in the lab packet
How to grind valves video	Video guide (link only)
Engine Assembly 1	Lab packet reflection + checklist (photo proof)
iCADEMY tech bits (engine check, rewind starter, valve clearance, chain/trimmer/blower/cut-off saw, chaps)	Slides 'Outdoor Power Equipment Service' + notes C (our own content)
Class slides	Deck: 17 slides, Words to Know first, hold-up-1-2-3 checks, recap
His worksheets (grouped into lessons)	Guided notes: 11 items + key
Class quiz / notes check	Quiz: 10 questions + key
Vocabulary	8 picture cards (cut apart)
—	1 game on the class site: 10 questions + a matching round

2. Supports built in

- Short lines, key words bold, word bank in every notes box
- Quiz: 3 choices, no "all of the above"
- Labeled parts tray (muffin tin) and a printed photo checklist at the bench
- Pair with a gen-ed partner; student owns specific steps
- Staff demonstrate each new tool first; student uses it with adult supervision

3. Add these as the IEP allows

- Read-aloud · extended time · separate setting
- Printed slides · videos paused at the "look for" part

4. What to grade

- Engine project: grade on the checklist steps completed with photo proof, not speed
- Troubleshooting: L1 points to fuel, spark, compression, air cards
- **Notes**: same points as his worksheets for that lesson when complete.
- **Quiz**: score, then scale to the class points.
- These are **modifications** (less content). The IEP must allow modified grading. Check with the case manager.

5. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm in his files say "check with the class teacher".

🚧 Under construction (check with the class teacher)

- **Engine project checklist and grading**: The project steps and rubric are done in class and not written in Canvas; our checklist is team-written.
- **Troubleshooting**: A syllabus goal with no class materials; our lessons are team-written.
- **iCADEMY tech bits**: STIHL iCADEMY is login-only; we wrote our own service content from public knowledge.

⚠️ Shop safety (with adult supervision)

- **Tools**: students may use all shop tools (hand tools, measuring tools, flywheel puller, valve tools, torque wrench, power tools) **with adult supervision**. Staff model first, stay at the bench, and count tools back in.
- **Eyes and body**: safety glasses on in the shop at all times; coveralls, closed-toe shoes, no loose clothing, long hair tied back, no dangling jewelry.
- **Fuel and oil**: gasoline is flammable. Drain fuel with adult supervision, outside or in a ventilated area, no sparks or flames nearby. Oily rags go in the metal safety can. Gloves for oil; wash hands after.

- **Exhaust**: engines are only started with adult supervision in a well-ventilated area or outdoors (carbon monoxide). Hot mufflers and cylinders burn: let parts cool.
- **Springs and sharp edges**: valve springs and rewind starter springs store energy; use the right tool with adult supervision and eyes protected. Blades, flywheel fins and piston rings can cut.
- **Before working on an engine**: disconnect the spark plug wire so the engine cannot start.
- **Lifting**: engines are heavy; lift with a partner, bend the knees. Car lifts, jacks and 18-wheeler work are video topics in this class.
- **Clean-up**: wipe spills right away (slips), sweep, return every tool to its spot.